

AYUSH KUMAR

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Education

Delhi Technological University

B.Tech in Engineering Physics

Aug. 2024 – Present

Delhi, India

St. Mary's School

Class XII – 88%

2023

Projects

Data Analyst Agent | *Python, Flask, Groq API, Llama 3.3 70B* | [GitHub](#) | [Live Demo](#) **May 2026 - Present**

- Built an autonomous AI-powered data analysis agent capable of understanding natural language dataset operations.
- Integrated Groq-hosted Llama 3.3 70B model for semantic query understanding and intelligent agent tool orchestration.
- Developed a modular agent-tool system for cleaning, querying, visualization, row operations, and CSV export.
- Combined LLM-driven reasoning with deterministic Pandas execution pipelines for reliable AI-assisted data processing.
- Designed conversational dataset interaction with real-time tables, charts, and persistent dataframe memory.

FitCheck | *Python, Flask, SQLAlchemy, Cloudinary* | [GitHub](#) | [Live Demo](#) **Apr. 2026 - May 2026**

- Built a full-stack fashion app for students to post outfits with brand, store, and price breakdowns.
- Implemented user authentication using bcrypt password hashing and Flask-Login to manage protected routes.
- Integrated Cloudinary to store images on the cloud and deployed the application on Render using Gunicorn.
- Designed trade and thrift database models allowing users to propose clothing swaps and list individual items for sale.
- Built thrift listing flow with item-level pricing so users can sell specific pieces from an outfit independently.

Movie Database with Semantic Search | *Flask, MongoDB Atlas, Python* | [GitHub](#) | [Live Demo](#) **May 2025**

- Built a full-stack movie rental app with CRUD operations using Flask, PyMongo, and MongoDB Atlas
- Implemented LLM-powered description generation via OpenRouter API to auto-enrich movie entries on insert
- Designed dual search — regex title matching and semantic vibe search using 384-dim vector embeddings
- Indexed 21,000+ movies with cosine similarity vectors in MongoDB Atlas for natural language querying

Bit Runner — 2D Side-Scrolling Game | *JavaScript, Phaser 3* | [GitHub](#) | [Live Demo](#) **July 2024**

- Built an interactive browser-based 2D side-scrolling game using JavaScript and the Phaser 3 game framework.
- Implemented player controls, collision detection, and sprite animations using JavaScript event handling techniques.
- Managed game state across the main menu, gameplay, and game over scenes using Phaser's scene architecture.

Open Source Contributions

Streamlit-webrtc | *Python, WebRTC* | [View PR](#) **Oct. 2025**

- Fixed WebRTC resource cleanup to prevent memory leaks and stabilize stream termination in a production library.
- Traced the stream termination flow across the codebase and patched the root cause without breaking existing tests.
- Pull request was reviewed and merged, demonstrating ability to contribute to real production open source codebases.

Opik | *Python, Data Visualization* | [View PR](#) **Oct. 2025**

- Implemented an in-app CSV preview feature improving how users inspect datasets before processing them.
- Built the UI component independently from reading the issue to writing code and submitting the pull request.
- Contribution improved the user-facing interface of a data platform used by machine learning engineers.

ACHIEVEMENTS

Video Hackathon

April 2025

2nd Runner Up

Delhi Technological University

- Rapidly prototyped a working video solution from scratch, demonstrating fast execution and creative thinking.
- Collaborated with team members to design, build, and present a complete project within the hackathon deadline.

Technical Skills

Languages: Python, C, JavaScript, SQL

Frontend: HTML, CSS

Backend: Node.js, Flask, REST API design

Databases: MongoDB, MySQL, SQL